DOUBLE JPEG COMPRESSION DETECTION FOR DISTINGUISHABLE BLOCKS IN IMAGES COMPRESSED WITH SAME QUANTIZATION MATRIX (Results)

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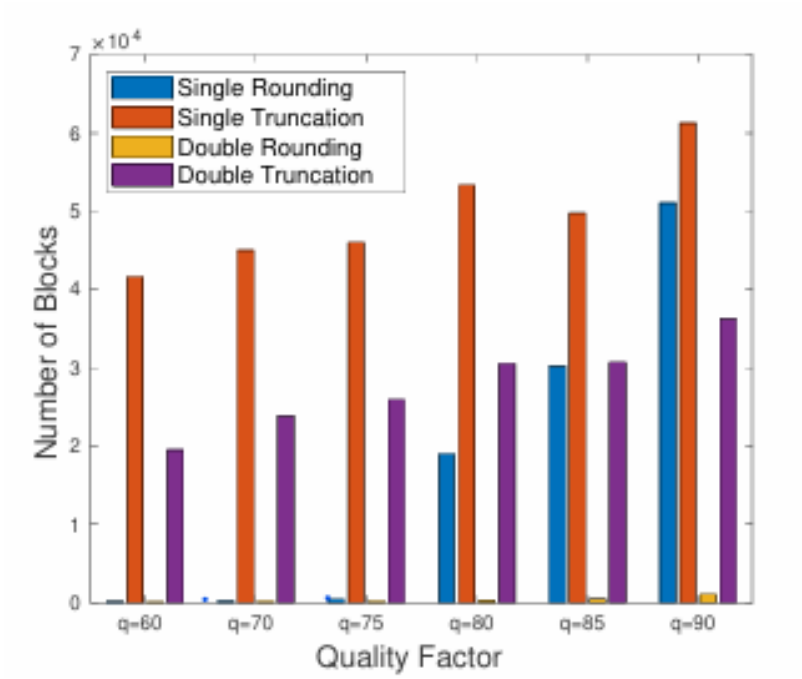
These are the results we are getting after the implementation -

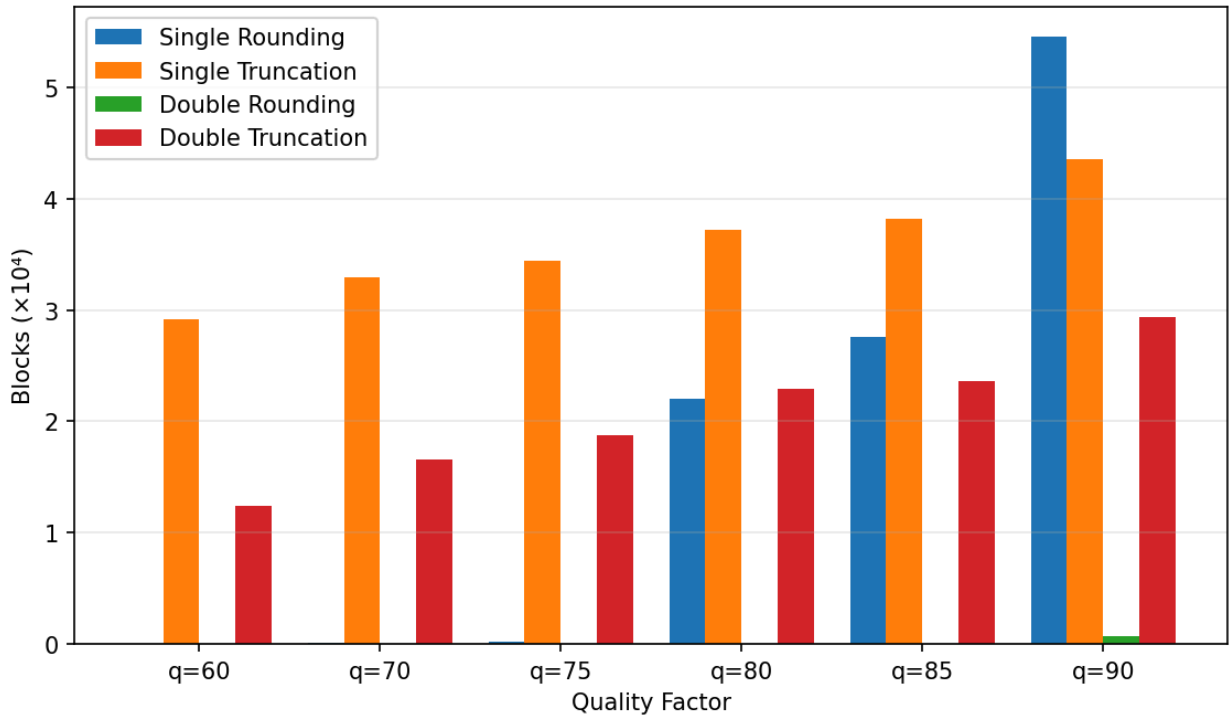
Rounding and Truncation Error Analysis

Expected



Our results

Fig 2. Rounding & Truncation Error Blocks (s ≥ 1)



Blocks with Stability Index s=1

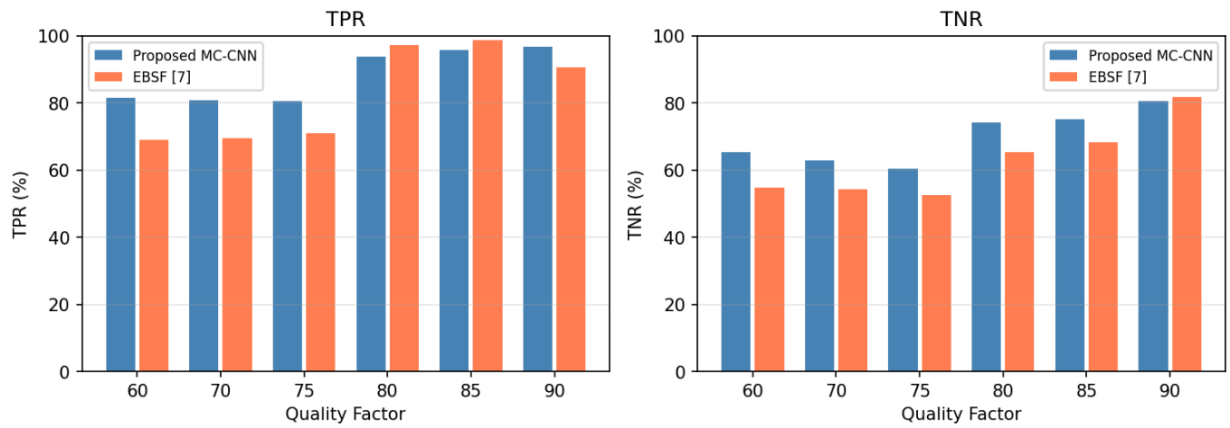
Expected

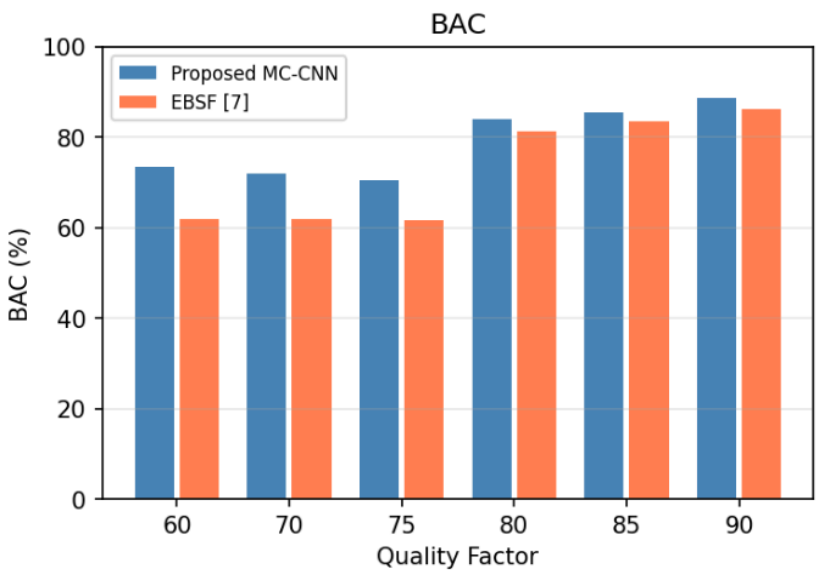
Table 2. Performance of the proposed approach and EBSF [7] based approach for the blocks with $s = 1$

Quality Factor	Proposed Approach			EBSF [7]		
	TPR	TNR	BAC	TPR	TNR	BAC
60	**85.89**	56.64	**71.26**	74.55	**62.10**	68.32
70	**82.99**	56.47	**69.73**	77.04	**56.51**	66.78
75	**80.76**	**56.74**	**68.75**	78.30	54.21	66.26
80	**95.71**	**68.43**	**82.07**	95.60	66.35	80.97
85	**96.69**	**71.84**	**84.27**	94.30	71.53	82.92
90	**97.44**	78.53	**87.98**	93.84	**79.10**	86.47

Our results

Table 2: Blocks with s = 1





Classification of Distinguishable Blocks

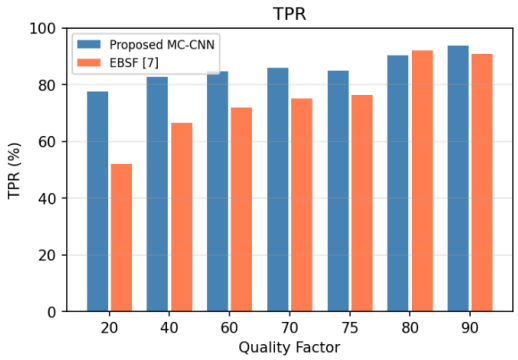
Expected

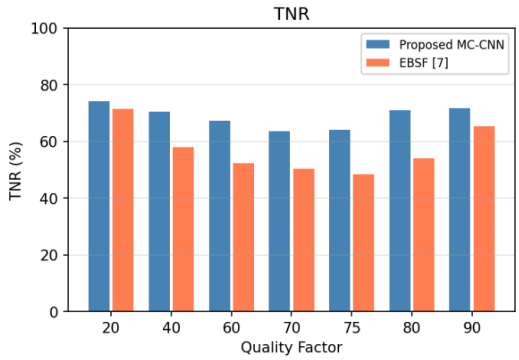
Table 3. Performance of the proposed approach and EBSF [7] based approach for all the distinguishable blocks

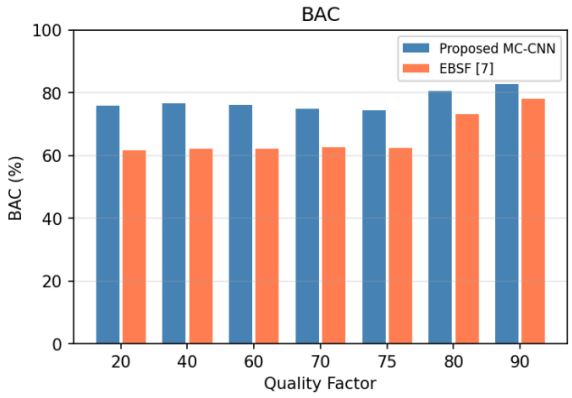
Quality Factor	Proposed Approach			EBSF [7]		
	TPR	TNR	BAC	TPR	TNR	BAC
20	**79.43**	59.00	69.21	78.04	**60.84**	**69.44**
40	**86.15**	**60.27**	**73.21**	81.45	58.49	69.97
60	**84.63**	62.15	**73.39**	76.07	**62.59**	69.33
70	**82.84**	**62.61**	**72.73**	78.22	60.17	69.20
75	**84.01**	**61.24**	**72.62**	79.24	58.02	68.63
80	**90.07**	**66.77**	**78.42**	88.88	62.00	75.44
85	**93.96**	**64.84**	**79.40**	89.39	64.16	76.78
90	**95.29**	**67.21**	**81.25**	92.81	65.17	78.99

Our results

Table 3: All Distinguishable Blocks (s ≥ 1)







Towards a Universal Detector

Expected

Table 4. Cross quality factor detection for distinguishable blocks

q_{train} \ q_{test}	60	70	75	80	85	90
60	**73.39**	72.60	71.82	76.52	75.95	69.63
70	73.23	**72.73**	72.44	75.10	74.14	66.71
75	73.32	72.70	**72.62**	68.56	69.83	61.55
80	70.81	69.66	68.98	**78.42**	79.09	75.20
85	68.98	67.46	66.67	77.76	**79.40**	78.29
90	61.71	60.65	60.24	74.69	77.35	**81.25**

Our results

Final results

QF	Proposed MC-CNN TPR	Proposed MC-CNN TNR	Proposed MC-CNN BAC	EBSF [7] TPR	EBSF [7] TNR	EBSF [7] BAC
60	81.46	65.43	73.45	69.02	54.76	61.89
70	80.80	62.99	71.89	69.65	54.38	62.02
75	80.50	60.56	70.53	71.00	52.56	61.78
80	93.86	74.12	83.99	97.41	65.27	81.34
85	95.90	75.08	85.49	98.68	68.34	83.51
90	96.77	80.51	88.64	90.67	81.78	86.23